

The *command line* is a text-based program that lets you enter commands to interact with the operating system and run programs. You might also hear it called the command line interface (CLI, which rhymes with “fly”), command prompt, terminal, shell, or console. It provides an alternative to a *graphical user interface* (GUI, pronounced “gooey”), which allows the user to interact with the computer through more than just a text-based interface. A GUI presents visual information to a user to guide them through tasks more easily than the command line does. Most computer users treat the command line as an advanced feature and never touch it. Part of the intimidation factor is due to the complete lack of hints of how to use it; although a GUI might display a button showing you where to click, a blank terminal window doesn’t remind you what to type.

But there are good reasons for becoming adept at using the command line. For one, setting up your environment often requires you to use the command line rather than the graphical windows. For another, entering commands can be much faster than clicking graphical windows with the mouse. Text-based commands are also less ambiguous than dragging an icon to some other icon. This lends them to automation better, because you can combine multiple specific commands into scripts to perform sophisticated operations.

The command line program exists in an executable file on your computer. In this context, we often call it a shell or shell program. Running the shell program makes the terminal window appear:

- On Windows, the shell program is at `C:\Windows\System32\cmd.exe`.
- On macOS, the shell program is at `/bin/bash`.
- On Ubuntu Linux, the shell program is at `/bin/bash`.

Over the years, programmers have created many shell programs for the Unix operating system, such as the Bourne Shell (in an executable file named *sh*) and later the Bourne-Again Shell (in an executable file named *Bash*). Linux uses Bash by default, whereas macOS uses the similar Zsh or Z shell in Catalina and later versions. Due to its different development history, Windows uses a shell named Command Prompt. All these programs do the same thing: they present a terminal window with a text-based CLI into which the user enters commands and runs programs.